
Key Derivations

A worked companion to the Soft Condensed Matter course

Part II Physics — University of Cambridge

The derivations that form the spine of the course, worked through in full with every intermediate step shown and the physical reasoning made explicit. They are organised into three tiers by importance: Tier 1 (foundational, examined most reliably), Tier 2 (important and rich in transferable technique), and Tier 3 (worth understanding, less often reproduced in full). Notation and framing follow the course lecture notes.

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1 Tier 1 — Must Know Cold

1.1 The Stokes drag, Einstein relation, and Stokes–Einstein

The idea

A colloid in a fluid feels two things from the same molecular collisions: a systematic drag that slows it down, and a random force that kicks it around. The fluctuation–dissipation theorem ties the two together; the result is that the diffusion constant is set by temperature divided by drag, and using the Stokes drag gives the celebrated $D = k_B T / 6\pi\eta r$.

Step 1: Stokes drag by a scaling argument. A sphere of radius a moving at velocity v acts as a steady source of transverse momentum into the fluid. In the low-Reynolds-number regime the flow obeys $\nabla^2 v_T = 0$ with no-slip at the surface; the solution that decays away from the sphere is

$$v_T(r) = v \frac{a}{r}.$$

The viscous stress on the surface is $\sigma' \sim \eta \partial_r v_T = \eta v a / r^2$, and integrating over the sphere's surface area $4\pi a^2$ (evaluated at $r = a$) gives

$$F \sim 4\pi a^2 \cdot \eta v \frac{a}{a^2} = 4\pi\eta a v.$$

The exact calculation only changes the prefactor:

Stokes drag

$$F = 6\pi\eta a v.$$

The key physics is $F \propto a$, not a^2 : drag is set by momentum *diffusing* away from the sphere, not by the cross-sectional area it sweeps.

Step 2: The Langevin equation. Separate the slow motion of the colloid from the fast solvent collisions. Newton's law with drag plus a random force $\vec{\xi}(t)$ is

$$m \frac{d\vec{v}}{dt} = -\gamma \vec{v} + \vec{\xi}(t), \quad \gamma = 6\pi\eta r,$$

where $\vec{\xi}$ is white noise:

$$\langle \vec{\xi}(t) \rangle = 0, \quad \langle \vec{\xi}(t) \cdot \vec{\xi}(t') \rangle = c \delta(t - t').$$

The delta-correlation says successive kicks (on timescales longer than the $\sim 10^{-13}$ s collision time) are statistically independent.

Step 3: Fix the noise strength by the fluctuation–dissipation theorem. At thermal equilibrium the equipartition theorem fixes the mean kinetic energy: $\frac{1}{2}m \langle v^2 \rangle = \frac{3}{2}k_B T$ in 3D. Solving the Langevin equation for $\langle v^2 \rangle$ and demanding consistency with equipartition forces the noise strength to be

$$c = 2\gamma k_B T \quad (\text{per dimension: } 6\gamma k_B T \text{ in 3D}).$$

This is the fluctuation–dissipation theorem: the same γ that *dissipates* energy also sets the size of the *fluctuating* force. They are two faces of one molecular process.

Step 4: Mean square displacement and the Einstein relation. Integrating the Langevin equation and averaging over the noise, the long-time ($t \gg m/\gamma$) mean square displacement grows linearly:

$$\langle \vec{x}(t)^2 \rangle = \frac{6k_B T}{\gamma} t \equiv 6Dt,$$

which identifies the diffusion coefficient (the Einstein relation)

$$D = \frac{k_B T}{\gamma}.$$

Step 5: Combine. Substituting the Stokes drag $\gamma = 6\pi\eta r$:

Stokes–Einstein relation

$$D = \frac{k_B T}{6\pi\eta r}.$$

The time for a colloid to diffuse its own size is $\tau_r = r^2/D = 6\pi\eta r^3/k_B T$.

Why it matters

This is the most-used single result in the course. It connects a *measurable* (the diffusion constant, e.g. from tracking a colloid) to molecular parameters, underpins microrheology, dynamic light scattering, and the whole Brownian-motion chapter. The deep content is the fluctuation–dissipation theorem: drag and noise are inseparable.

1.2 The ideal chain: end-to-end distance and the entropic spring

The idea

Treat a polymer as a random walk of N independent steps. The end-to-end vector is then Gaussian-distributed, exactly like a diffusing particle's displacement. Because stretching the chain reduces the number of available conformations, it lowers the entropy — so the chain resists stretching with a purely entropic, spring-like force.

Step 1: Mean square end-to-end distance. The end-to-end vector is the sum of N bond vectors, $\vec{R} = \sum_i \vec{u}_i$, each of length b_0 . Then

$$\langle \vec{R}^2 \rangle = \sum_{i,j} \langle \vec{u}_i \cdot \vec{u}_j \rangle = \sum_i \langle u_i^2 \rangle + \sum_{i \neq j} \langle \vec{u}_i \cdot \vec{u}_j \rangle.$$

For an ideal chain the bonds are uncorrelated, so the cross terms vanish ($\langle \vec{u}_i \cdot \vec{u}_j \rangle = 0$ for $i \neq j$), leaving

$$\langle \vec{R}^2 \rangle = N b_0^2.$$

The chain size scales as $\sqrt{N}$, far smaller than the contour length $N b_0$ — the random walk doubles back on itself.

Step 2: The Gaussian distribution. Following the chain from one end to the other is a 3D random walk, so by the central limit theorem the end-to-end vector is Gaussian. In each dimension the variance is $\langle R_x^2 \rangle = N b_0^2/3$, giving

$$P(\vec{R}) = \left(\frac{3}{2\pi N b_0^2} \right)^{3/2} \exp\left(-\frac{3\vec{R}^2}{2N b_0^2} \right).$$

The probability is maximal at $\vec{R} = 0$ (the coiled state) and falls off for stretched configurations.

Step 3: Entropy and free energy. The entropy of a chain constrained to end-to-end vector $\vec{R}$ is $S = k_B \ln P(\vec{R})$, so

$$S(\vec{R}) = -k_B \frac{3\vec{R}^2}{2N b_0^2} + S(0).$$

An ideal chain has no internal energy ($U = 0$, since segments don't interact), so the Helmholtz free energy $F = U - TS$ is

Free energy of an ideal chain

$$F(\vec{R}) = \frac{3k_B T}{2Nb_0^2} \vec{R}^2 + F(0).$$

Step 4: The entropic spring. The force needed to hold the ends at separation $\vec{R}$ is the gradient of the free energy:

$$\vec{f} = \frac{\partial F}{\partial \vec{R}} = \frac{3k_B T}{Nb_0^2} \vec{R}.$$

This is Hooke's law with an effective spring constant

$$k_{\text{spring}} = \frac{3k_B T}{Nb_0^2}.$$

Why it matters

The spring constant is proportional to temperature — a hotter entropic spring is *stiffer*, the opposite of an energetic (bond) spring. This is the molecular origin of rubber elasticity: heat a stretched rubber band and it pulls in harder. The Gaussian chain is the polymer analogue of the diffusing Brownian particle, and the same $F \propto R^2$ underlies rubber elasticity, polymer solution thermodynamics, and the worm-like chain.

1.3 The Flory–Huggins free energy and the critical point

The idea

Put polymer and solvent on a lattice. Mixing gains entropy (more ways to arrange the molecules) but costs energy (unfavourable polymer–solvent contacts). The competition gives a free energy whose curvature decides whether the solution stays mixed or separates into two phases — and the polymer's large size makes it far easier to demix than a small molecule.

Step 1: Entropy of mixing. On a lattice of $M = N_A + N_B$ sites, the number of ways to arrange N_A solute and N_B solvent molecules is $\Omega = M!/(N_A!N_B!)$. Using Stirling's approximation $\ln M! \approx M \ln M - M$, the entropy of mixing per site, in terms of volume fraction $\Phi = N_A/M$, is

$$\Delta S = -k_B M [\Phi \ln \Phi + (1 - \Phi) \ln(1 - \Phi)].$$

For a *polymer* of N segments that must occupy adjacent sites, the translational entropy is reduced by a factor N (the N segments move together, not independently):

$$\Delta S = -k_B M \left[\frac{\Phi}{N} \ln \Phi + (1 - \Phi) \ln(1 - \Phi) \right].$$

This single factor of $1/N$ is why polymers are so much harder to dissolve than small molecules.

Step 2: Energy of mixing. With nearest-neighbour interaction energies $\epsilon_{pp}, \epsilon_{ps}, \epsilon_{ss}$ and coordination number z , a mean-field count of contacts gives the energy change on mixing

$$\Delta U = \frac{zM}{2} \Phi(1 - \Phi) [2\epsilon_{ps} - \epsilon_{ss} - \epsilon_{pp}] \equiv \chi M k_B T \Phi(1 - \Phi),$$

defining the Flory interaction parameter

$$\chi = \frac{z}{2} \frac{2\epsilon_{ps} - \epsilon_{ss} - \epsilon_{pp}}{k_B T}.$$

$\chi > 0$ means polymer–solvent contacts are unfavourable.

Step 3: The free energy of mixing. Combining $f = \Delta U/M - T\Delta S/M$:

Flory–Huggins free energy density

$$\frac{f_{\text{mix}}}{k_B T} = \frac{\Phi}{N} \ln \Phi + (1 - \Phi) \ln(1 - \Phi) + \chi \Phi(1 - \Phi).$$

The logarithmic terms (entropy) favour mixing; the χ term (energy) opposes it when $\chi > 0$.

Step 4: The critical point. Phase separation becomes possible when f_{mix} develops regions of negative curvature ($\partial^2 f / \partial \Phi^2 < 0$, the spinodal). The critical point is the spinodal's extremum, $\partial^2 f / \partial \Phi^2 = \partial^3 f / \partial \Phi^3 = 0$. Differentiating twice,

$$\frac{\partial^2 (f/k_B T)}{\partial \Phi^2} = \frac{1}{N\Phi} + \frac{1}{1 - \Phi} - 2\chi,$$

and setting the derivative of the spinodal condition to zero gives

$$\Phi_c = \frac{1}{\sqrt{N} + 1}, \quad \chi_c = \frac{1}{2} + \frac{1}{\sqrt{N}} + \frac{1}{2N} \approx \frac{1}{2} + \frac{1}{\sqrt{N}}.$$

Why it matters

For small molecules ($N = 1$) the critical point is at $\chi_c = 2$ and $\Phi_c = \frac{1}{2}$ — symmetric demixing. For a long polymer ($N \rightarrow \infty$), $\chi_c \rightarrow \frac{1}{2}$ and $\Phi_c \rightarrow 0$: a polymer phase -separates from solvent at much weaker interactions, and at tiny polymer fractions. The value $\chi = \frac{1}{2}$ is the famous theta point, where the chain behaves ideally. This derivation governs polymer solubility, the coil–globule transition, and the phase diagrams of the course.

1.4 Young's equation for the contact angle

The idea

A droplet sitting on a solid meets it along a contact line where three interfaces — solid–liquid, liquid–vapour, solid–vapour — come together. Each interface pulls on that line with a force per unit length equal to its surface tension. Balancing the horizontal pull gives the equilibrium contact angle.

Step 1: The three surface tensions. At the contact line, three surface energies (energies per unit area, or equivalently forces per unit length) act: γ_{SL} (solid–liquid), γ_{LV} (liquid–vapour), and γ_{SV} (solid–vapour). The droplet meets the solid at the contact angle Θ , measured inside the liquid.

Step 2: Balance forces along the surface. Resolving the tensions horizontally (along the solid surface) at the contact line: the solid–vapour tension pulls outward, opposed by the solid–liquid tension plus the horizontal component of the liquid–vapour tension, $\gamma_{LV} \cos \Theta$:

Young's equation

$$\gamma_{SL} + \gamma_{LV} \cos \Theta = \gamma_{SV}.$$

Equivalently the equilibrium contact angle is

$$\cos \Theta = \frac{\gamma_{SV} - \gamma_{SL}}{\gamma_{LV}}.$$

Step 3: Wetting regimes and the spreading parameter. Define the spreading parameter $S = \gamma_{SV} - (\gamma_{SL} + \gamma_{LV})$, the energy difference between a dry and a wet substrate:

- $S > 0$: the right-hand side of $\cos \Theta$ exceeds 1, there is no real solution, and the liquid spreads completely ($\Theta = 0$) — *total wetting*.
- $-2\gamma_{LV} < S < 0$: a finite contact angle, *partial wetting*. $\Theta < 90^\circ$ is hydrophilic, $\Theta > 90^\circ$ hydrophobic.
- $\cos \Theta = -1$: $\Theta = 180^\circ$, complete *dewetting*.

Why it matters

Young's equation is a force balance *along* the surface; the vertical component is taken up by elastic deformation of the solid and is usually ignored. The same tensions, applied to a curved interface, give the Laplace pressure and capillary rise. This is the foundation of the wetting and capillarity material.

2 Tier 2 — Important, Technique-Rich

2.1 Poiseuille flow in a channel

The idea

A pressure difference drives viscous fluid down a channel. With no acceleration (steady, low-Reynolds flow), the pressure gradient is balanced locally by viscous shear. Integrating that balance with no-slip walls gives a parabolic velocity profile, and integrating the profile gives the flow rate — which scales dramatically with channel size.

Step 1: Force balance. For steady, unidirectional, low-Reynolds flow $v_x(z)$ driven by a pressure drop Δp over length L , the Navier–Stokes equation reduces to a balance between the pressure gradient and viscous stress:

$$\eta \partial_z^2 v_x(z) = -\frac{\Delta p}{L}.$$

Step 2: Parallel plates. With no-slip walls at $z = 0$ and $z = h$, integrate twice:

$$v_x(z) = \frac{\Delta p}{2\eta L} (h - z)z,$$

a parabolic profile, maximal at the centre and zero at both walls. The flow rate through a width w is

$$Q = \int_0^w dy \int_0^h dz v_x(z) = \frac{h^3 w}{12\eta L} \Delta p.$$

Step 3: Circular pipe. The same balance in cylindrical coordinates, $\frac{\eta}{r} \frac{d}{dr} \left(r \frac{dv_x}{dr} \right) = -\Delta p/L$, with no-slip at $r = R$ gives the parabolic profile $v_x(r) = \frac{\Delta p}{4\eta L} (R^2 - r^2)$. Integrating over the cross-section,

Hagen–Poiseuille law

$$Q = \int_0^R v_x(r) 2\pi r dr = \frac{\pi R^4}{8\eta L} \Delta p.$$

Step 4: Hydraulic resistance. Writing $\Delta p = Q R_{\text{hyd}}$ in analogy with Ohm’s law gives the hydraulic resistance $R_{\text{hyd}} = 8\eta L/\pi R^4$.

Why it matters

The R^4 dependence is the headline: halving a pipe’s radius raises its resistance sixteenfold. This is why microfluidic channels have enormous resistance, why blood-vessel constriction matters so much, and why the hydraulic–electrical analogy (resistances in series/parallel) is so useful for circuit-like channel networks.

2.2 Freely-jointed chain: the force–extension law

The idea

Model a polymer under tension as N independent segments, each either pointing along the force or against it — a row of two-state spins in a field. The statistical mechanics is identical to a paramagnet, and the average extension follows the same tanh law, capping at the contour length.

Step 1: The two-state model. Each segment i has projection $\sigma_i = \pm 1$ along the applied force f , contributing $\pm a$ to the extension. The total extension is $z = a \sum_{i=1}^N \sigma_i$, and the energy of a segment is $-fa\sigma_i$ (aligned segments are lower in energy).

Step 2: Single-segment average. Each segment is independent, so it is a two-level system with Boltzmann weights $e^{\pm fa/k_B T}$. The mean projection is

$$\langle \sigma_i \rangle = \frac{(+1)e^{fa/k_B T} + (-1)e^{-fa/k_B T}}{e^{fa/k_B T} + e^{-fa/k_B T}} = \tanh\left(\frac{fa}{k_B T}\right).$$

Step 3: Total extension. Summing the N independent segments, with contour length $L = Na$,

Freely-jointed chain force–extension

$$\langle z \rangle = Na \tanh\left(\frac{fa}{k_B T}\right) = L \tanh\left(\frac{fa}{k_B T}\right).$$

Step 4: Limits. At low force, $\tanh x \approx x$, so $\langle z \rangle \approx L fa/k_B T = Na^2 f/k_B T$ — a linear (Hookean) entropic spring with $k = k_B T/(Na^2)$, matching the Gaussian-chain result. At high force, $\tanh x \rightarrow 1$, so $\langle z \rangle \rightarrow L$: the extension saturates at the contour length, as it must for a finite chain.

Why it matters

This is the polymer version of paramagnetic (Langevin/Brillouin) saturation. The low-force linear regime recovers the entropic spring; the high-force plateau is the geometric limit. Real single-molecule force spectroscopy (optical/magnetic tweezers on DNA) is fit with exactly this form (or its worm-like-chain refinement).

2.3 Debye–Hückel screening length

The idea

A charged surface in an electrolyte attracts counter-ions, which cloud around it and screen its field. Combining electrostatics (Poisson) with thermal equilibrium of the ions (Boltzmann) gives a nonlinear equation; linearising it for weak potentials yields exponential screening over a characteristic length set by the salt concentration.

Step 1: Poisson plus Boltzmann. The potential obeys Poisson's equation $\nabla^2 \phi = -\rho/\epsilon$, with charge density $\rho = \sum_i e z_i n_i$. In equilibrium the ion densities follow the Boltzmann distribution $n_i = n_{i0} \exp(-e z_i \phi/k_B T)$. Combining gives the *Poisson–Boltzmann equation*

$$\nabla^2 \phi = -\frac{e}{\epsilon} \sum_i z_i n_{i0} \exp\left(-\frac{e z_i \phi}{k_B T}\right).$$

Step 2: Linearise for weak potentials. When $e\phi \ll k_B T$, expand the exponential to first order, $\exp(-e z_i \phi/k_B T) \approx 1 - e z_i \phi/k_B T$:

$$\nabla^2 \phi \approx -\frac{e}{\epsilon} \sum_i z_i n_{i0} + \frac{e^2}{\epsilon k_B T} \sum_i z_i^2 n_{i0} \phi.$$

The first term vanishes by bulk charge neutrality ($\sum_i z_i n_{i0} = 0$), leaving the *Debye–Hückel equation*

$$\nabla^2 \phi = \kappa^2 \phi.$$

Step 3: The screening length.

Debye screening length

$$\kappa^2 = \frac{e^2}{\epsilon k_B T} \sum_i z_i^2 n_{i0}, \quad \lambda_D = \frac{1}{\kappa}.$$

In the planar case the solution is $\phi(x) = \phi_0 e^{-\kappa x}$: the potential decays exponentially, falling by $1/e$ over one Debye length.

Step 4: Salt dependence. Because $\kappa^2 \propto \sum_i z_i^2 n_{i0}$, adding salt increases κ and *shrinks* λ_D : more ions screen more effectively. Multivalent ions (z_i^2) are disproportionately effective. Adding salt also improves the validity of the linearisation (smaller potentials).

Why it matters

The Debye length is the single most important length in colloid science: it sets the range of the screened-Coulomb repulsion in DLVO theory, hence colloidal stability. Unlike the Bjerrum length (fixed by the solvent), λ_D is tunable by salt — which is exactly how one “salts out” a colloidal suspension to make it aggregate.

2.4 Rubber elasticity: stress of a deformed network

The idea

A rubber is a network of cross-linked entropic springs. Assume each strand deforms in proportion to the bulk (the affine assumption). Summing the entropic free-energy change of all strands gives the network free energy, and differentiating gives a stress-strain law with a purely entropic, temperature-proportional modulus.

Step 1: Affine deformation. Under a deformation with stretch ratios $\lambda_x, \lambda_y, \lambda_z$, each strand's end-to-end vector transforms as $R_x = \lambda_x R_{x0}$, etc. Using the ideal-chain free energy $F = \frac{3k_B T}{2Nb_0^2} \vec{R}^2$, the change per strand is

$$\Delta F = \frac{3k_B T}{2Nb_0^2} [(\lambda_x^2 - 1)R_{x0}^2 + (\lambda_y^2 - 1)R_{y0}^2 + (\lambda_z^2 - 1)R_{z0}^2].$$

Step 2: Average over strands. For n strands crosslinked in their ideal state, $\sum_i (R_{x0}^2)_i = n \langle R_{x0}^2 \rangle = n N b_0^2 / 3$ (isotropy), and similarly for y, z . Summing,

$$\Delta F = \frac{nk_B T}{2} (\lambda_x^2 + \lambda_y^2 + \lambda_z^2 - 3).$$

Notice N and b_0 have cancelled — the network free energy depends only on the number of strands and temperature.

Step 3: Incompressible uniaxial stretch. Rubber conserves volume, so $\lambda_x \lambda_y \lambda_z = 1$. For a uniaxial stretch $\lambda_x = \lambda$, symmetry gives $\lambda_y = \lambda_z = 1/\sqrt{\lambda}$, so

$$\Delta F = \frac{nk_B T}{2} \left(\lambda^2 + \frac{2}{\lambda} - 3 \right).$$

Step 4: Stress. The force is $\partial F / \partial L_x$; dividing by area and using the strand number density $\nu = n/V$ gives the engineering stress

Rubber stress-strain law

$$\sigma = \nu k_B T \left(\lambda - \frac{1}{\lambda^2} \right), \quad \text{small strain: } \sigma \approx 3\nu k_B T \varepsilon,$$

so the shear modulus is $G = \nu k_B T$, proportional to temperature and to crosslink density.

Why it matters

The modulus $G = \nu k_{\text{B}} T$ rising with temperature is the signature of entropic elasticity: a stretched rubber band heated up contracts more strongly, and warms when stretched adiabatically (the Gough–Joule effect). The nonlinear $\lambda - \lambda^{-2}$ form (not Hookean) is characteristic of large rubber deformations.

3 Tier 3 — Worth Understanding

3.1 Van der Waals attraction between two plates (Hamaker)

The idea

Two neutral atoms attract via a $-1/r^6$ dispersion potential. To get the force between two macroscopic bodies, add up the pairwise interactions of all their atoms. For two semi-infinite plates the sum (integral) converts the steep $1/r^6$ atomic law into a much longer-ranged $1/h^2$ surface law.

Step 1: The pairwise potential. Two atoms a distance r apart attract with $u(r) = -L/r^6$, where L sets the dispersion strength. The plates have atomic number density n .

Step 2: Atom-to-plate integral. First find the energy of one atom at distance D from the surface of a semi-infinite solid. Integrate $-L/r^6$ over the half-space, using cylindrical shells. The result of the integral scales as

$$u_{\text{atom-plate}}(D) = -\frac{\pi n L}{6D^3}.$$

The $1/r^6$ pair law has become a $1/D^3$ law after integrating over one body.

Step 3: Plate-to-plate integral. Now integrate this over all atoms (density n) in the second plate, which occupies D from h to ∞ . The energy per unit area is

$$W(h) = -n \int_h^\infty \frac{\pi n L}{6D^3} dD = -\frac{\pi n^2 L}{12h^2}.$$

Van der Waals energy between plates

$$W(h) = -\frac{A}{12\pi h^2}, \quad A = \pi^2 n^2 L,$$

where A is the Hamaker constant.

Step 4: Range. Each integration over a body raises the power of the distance law: $r^{-6} \rightarrow D^{-3} \rightarrow h^{-2}$. The macroscopic interaction is thus far longer-ranged than the atomic one.

Why it matters

The Hamaker constant A bundles all the material-specific dispersion physics into one number ($\sim 10^{-20}$ J for most condensed matter). The $1/h^2$ attraction is the attractive well in DLVO theory; it competes with the screened-Coulomb repulsion to decide colloidal stability.

3.2 Capillary rise (Jurin's law)

The idea

A wetting liquid in a thin tube curves its surface into a meniscus. The curvature creates a Laplace pressure drop, which the liquid column's weight must balance. Equating the two gives the equilibrium rise height, inversely proportional to the tube radius.

Step 1: Laplace pressure at the meniscus. A curved interface with principal radii R_1, R_2 supports a pressure jump $\Delta p = \gamma(1/R_1 + 1/R_2)$. For a rotationally symmetric capillary of radius R with a hemispherical wetting meniscus, $R_1 = R_2 = R/\cos \Theta$, so $\Delta p = 2\gamma \cos \Theta/R$.

Step 2: Balance against hydrostatic pressure. The raised column of height h exerts hydrostatic pressure ρgh . At equilibrium this balances the Laplace pressure:

$$\frac{2\gamma \cos \Theta}{R} = \rho gh.$$

Step 3: The rise height.**Jurin's law**

$$h = \frac{2\gamma \cos \Theta}{\rho g R}.$$

For complete wetting ($\Theta = 0$), $h = 2\gamma/\rho g R$. The capillary length $\ell_c = \sqrt{\gamma/\rho g}$ (about 2.7 mm for water) sets the scale below which surface tension dominates gravity.

Step 4: Dynamics (Washburn). Before equilibrium, the rise is resisted by Poiseuille drag in the tube. Balancing capillary driving against viscous resistance gives a diffusive-looking growth $L(t) = \sqrt{R\gamma \cos \Theta t/2\eta}$ — the $L \sim \sqrt{t}$ Washburn law (problem-sheet Q17b).

Why it matters

Jurin's law (static) measures surface tension; the Washburn law (dynamic) governs wicking into porous media, paper, and soil. Both follow from the same Laplace-pressure source, balanced against gravity or viscous drag respectively.

3.3 Micelle shape and the packing parameter**The idea**

A surfactant has a head of fixed area and a tail of fixed volume and length. Whether the molecules pack into spheres, cylinders, or bilayers is decided by a single dimensionless ratio of these three geometric quantities — the packing parameter.

Step 1: Geometry of a spherical micelle. For α surfactants of tail volume v and head area a_0 forming a sphere of radius R : the volume condition is $V = \frac{4}{3}\pi R^3 = \alpha v$, and the surface condition is $A = 4\pi R^2 = \alpha a_0$. Dividing the two,

$$R = \frac{3v}{a_0}.$$

Step 2: The packing constraint. The core must be filled by the tails, so the radius cannot exceed the tail length: $R < \ell$ (the critical chain length). Substituting $R = 3v/a_0$:

$$\frac{3v}{a_0} < \ell \quad \implies \quad \frac{v}{a_0 \ell} < \frac{1}{3}.$$

Packing parameter

$$p = \frac{v}{a_0 \ell}, \quad \begin{cases} p < \frac{1}{3} & \text{spherical micelles} \\ \frac{1}{3} < p < \frac{1}{2} & \text{cylindrical micelles} \\ \frac{1}{2} < p < 1 & \text{bilayers / vesicles} \\ p \approx 1 & \text{planar bilayers} \\ p > 1 & \text{inverted structures} \end{cases}$$

Step 3: Physical reading. Small p means a bulky head and skinny tail — a cone that packs into a high-curvature sphere. As $p \rightarrow 1$ the molecule becomes cylindrical and packs into flat bilayers. The same geometric reasoning for cylinders ($V = \pi R^2 \ell$, $A = 2\pi R \ell$) gives the $p < \frac{1}{2}$ bound, and for bilayers the $p < 1$ bound (problem-sheet Q30b).

Why it matters

The packing parameter predicts self-assembled morphology from three single-molecule measurements, with no free parameters. It explains why single-tailed surfactants (small v) make micelles while double-tailed lipids (large v) make the bilayers of cell membranes.

3.4 Membrane fluctuation spectrum (Helfrich)**The idea**

A membrane resists bending and tension. Expand its shape into Fourier modes; the energy becomes a sum of independent harmonic modes. Equipartition then assigns $\frac{1}{2}k_B T$ to each, giving the mean-square amplitude of every wavelength — the fluctuation spectrum.

Step 1: The energy functional. A nearly flat membrane with height profile $h(\vec{r})$, tension σ and bending rigidity κ has energy

$$E = \int d^2r \left[\frac{\sigma}{2} (\nabla h)^2 + \frac{\kappa}{2} (\nabla^2 h)^2 \right].$$

The first term penalises stretching the area; the second penalises curvature.

Step 2: Fourier expand. Write $h(\vec{r}) = \frac{A}{(2\pi)^2} \int d^2q \tilde{h}_{\vec{q}} e^{i\vec{q}\cdot\vec{r}}$. Each gradient brings down a factor of $i\vec{q}$ and each Laplacian a factor of $-q^2$, so $(\nabla h)^2 \rightarrow q^2 |\tilde{h}_{\vec{q}}|^2$ and $(\nabla^2 h)^2 \rightarrow q^4 |\tilde{h}_{\vec{q}}|^2$. The energy becomes a sum of independent modes:

$$E = \frac{A}{2} \int \frac{d^2q}{(2\pi)^2} [\sigma q^2 + \kappa q^4] |\tilde{h}_{\vec{q}}|^2.$$

Each mode $\vec{q}$ is an independent harmonic degree of freedom.

Step 3: Equipartition. Each quadratic mode gets mean energy $\frac{1}{2}k_B T$:

$$\frac{1}{2}k_B T = \frac{A}{2} [\sigma q^2 + \kappa q^4] \langle |\tilde{h}_{\vec{q}}|^2 \rangle.$$

Solving,

Membrane fluctuation spectrum

$$\langle |\tilde{h}_{\vec{q}}|^2 \rangle = \frac{k_B T}{A} \frac{1}{\sigma q^2 + \kappa q^4}.$$

Step 4: Two regimes. The crossover wavevector is $q^* = \sqrt{\sigma/\kappa}$. For $q < q^*$ (long wavelengths) tension dominates and $\langle |\tilde{h}_{\vec{q}}|^2 \rangle \sim k_B T / (A \sigma q^2)$; for $q > q^*$ (short wavelengths) bending dominates and $\langle |\tilde{h}_{\vec{q}}|^2 \rangle \sim k_B T / (A \kappa q^4)$. Summing over modes gives the real-space roughness $\langle h^2 \rangle$ (problem-sheet Q33).

Why it matters

This is the soft-matter analogue of the Rayleigh–Jeans counting of modes: equipartition over Fourier modes of a quadratic energy. It governs membrane undulations, the steric (Helfrich) repulsion between membranes, and is measured by flicker spectroscopy on red blood cells. The same machinery (Fourier modes + equipartition) gives capillary-wave roughness of liquid interfaces.

End of derivations companion. Each derivation is self-contained; read it alongside the corresponding chapter of the course lecture notes for the surrounding context and figures. Notation follows the course handout.